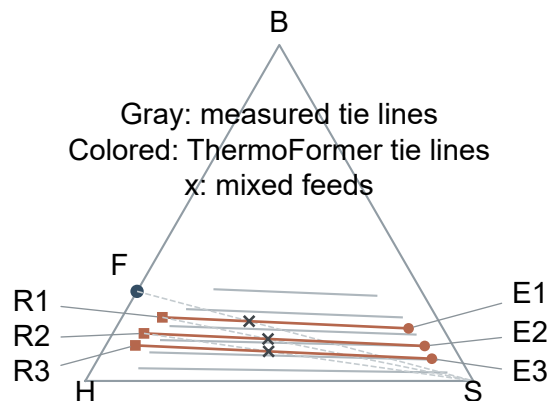


Three-stage crossflow extraction of benzene with furfural

Feed: 1 mol s⁻¹; $z_H = 0.733333$, $z_B = 0.266667$ | $P = 101.3$ kPa | Main design: 303.15 K, 0.5 mol s⁻¹ furfural per stage

a Equilibrium path at 303.15 K



b Experimental tie-line endpoints (mole fractions)

T (K)	Line	E : x_H	E : x_B	E : x_S	R : x_H	R : x_B	R : x_S
298.15	1	0.0544	0.0242	0.9214	0.9069	0.0291	0.0640
298.15	2	0.0639	0.0630	0.8731	0.8509	0.0759	0.0732
298.15	3	0.0715	0.1030	0.8255	0.7984	0.1197	0.0819
298.15	4	0.0754	0.1382	0.7864	0.7496	0.1618	0.0886
298.15	5	0.0939	0.1868	0.7193	0.7081	0.1961	0.0958
298.15	6	0.1307	0.2440	0.6253	0.6031	0.2626	0.1343
303.15	1	0.0569	0.0251	0.9180	0.8418	0.0377	0.1205
303.15	2	0.0618	0.0631	0.8751	0.7897	0.0850	0.1253
303.15	3	0.0715	0.1035	0.8250	0.7440	0.1215	0.1345
303.15	4	0.0799	0.1380	0.7821	0.6987	0.1632	0.1381
303.15	5	0.0897	0.1892	0.7211	0.6308	0.2136	0.1556
303.15	6	0.1232	0.2536	0.6232	0.5306	0.2728	0.1966

c Complete outlet streams: three-stage main design

ThermoFormer					Experimental reference				UNIFAC-LLE			
Stream	Flow	x_H	x_B	x_S	Flow	x_H	x_B	x_S	Flow	x_H	x_B	x_S
E1	0.528718	0.088566	0.156502	0.754932	0.461572	0.084221	0.160576	0.755202	0.539512	0.088863	0.169895	0.741242
R1	0.971282	0.706805	0.189359	0.103836	1.038428	0.668760	0.185424	0.145817	0.960488	0.713586	0.182205	0.104208
E2	0.650159	0.073225	0.104101	0.822674	0.654293	0.073960	0.113603	0.812437	0.653534	0.073783	0.114177	0.812040
R2	0.821123	0.778080	0.141561	0.080360	0.884135	0.730734	0.133712	0.135554	0.806953	0.789601	0.124403	0.085996
E3	0.593727	0.073770	0.066204	0.860026	0.603392	0.064328	0.073631	0.862041	0.598798	0.064702	0.072760	0.862538
R3	0.727397	0.818123	0.105763	0.076114	0.780743	0.777788	0.094514	0.127698	0.708155	0.845051	0.080235	0.074714

All flows: mol s⁻¹. H: *n*-heptane; B: benzene; S: furfural. Each source independently carries its full raffinate composition into the next stage.

d Overall results: three-stage main design

Quantity	ThermoFormer	Reference	UNIFAC-LLE
Benzene recovery (%)	71.15	72.33	78.69
Heptane retention (%)	81.15	82.81	81.60
Final raffinate flow	0.727397	0.780743	0.708155
Combined extract flow	1.772603	1.719257	1.791845
Furfural flow in raffinate	0.055365	0.099699	0.052909
Heptane in raffinate (mol%)	81.81	77.78	84.51
Benzene in raffinate (mol%)	10.58	9.45	8.02
Furfural in raffinate (mol%)	7.61	12.77	7.47
Benzene in extract (mol%)	10.70	11.22	11.71
Furfural in extract (mol%)	81.50	81.45	80.76
Solvent-free raffinate H (mol%)	88.55	89.16	91.33
Solvent-free extract B (mol%)	57.85	60.47	60.87

e Stage selection: fixed total solvent flow

B recovery (%)					H retention (%)		
T (K)	N	1	2	3	1	2	3
298.15	1	61.63	66.24	67.65	82.92	83.11	83.70
298.15	2	68.44	73.25	75.42	81.76	82.71	82.73
298.15	3	71.24	77.14	78.60	81.17	81.85	82.31
298.15	4	72.75	78.84	80.32	80.86	81.64	82.08
303.15	1	61.56	63.32	67.78	82.89	83.80	83.05
303.15	2	68.35	69.45	75.53	81.73	83.07	82.04
303.15	3	71.15	72.33	78.69	81.15	82.81	81.60
303.15	4	72.65	73.84	80.41	80.84	82.63	81.36

Targets: B recovery $\geq 70\%$; H retention $\geq 80\%$.
1 ThermoFormer; 2 experimental reference; 3 UNIFAC-LLE.
Total furfural: 1.5 mol s⁻¹, divided equally.

f Best recovery in discrete allocation scans

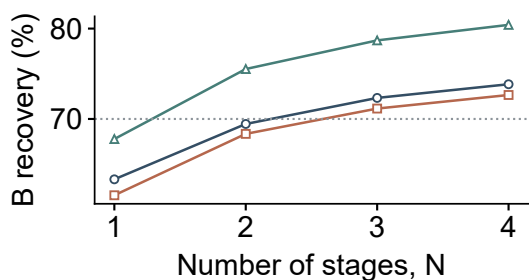
N	Source	Stagewise solvent	B rec. (%)	H ret. (%)
2	ThermoFormer	0.80 / 0.70	68.38	81.82
2	Reference	0.90 / 0.60	69.67	83.04
2	UNIFAC-LLE	0.80 / 0.70	75.59	82.05
3	ThermoFormer	0.50 / 0.50 / 0.50	71.15	81.15
3	Reference	0.50 / 0.50 / 0.50	72.33	82.81
3	UNIFAC-LLE	0.50 / 0.50 / 0.50	78.69	81.60

g Reference interpolation sensitivity

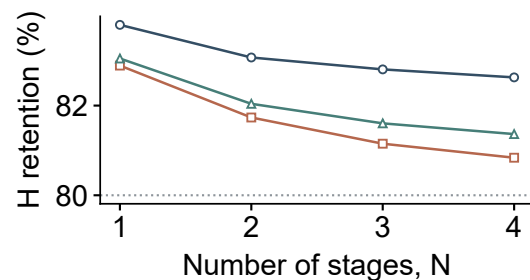
T (K)	N	Linear (%)	PCHIP (%)	Diff. (pp)
298.15	2	73.25	73.18	-0.071
298.15	3	77.14	77.10	-0.041
303.15	2	69.45	69.41	-0.039
303.15	3	72.33	72.26	-0.066

Benzene recovery; differences before rounding.
Interpolation is restricted to measured tie lines.

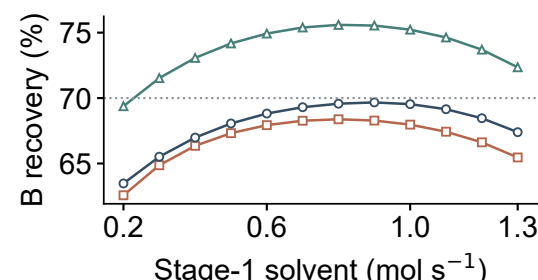
h Stage count: B recovery



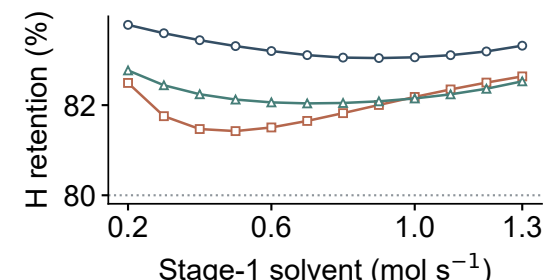
i Stage count: H retention



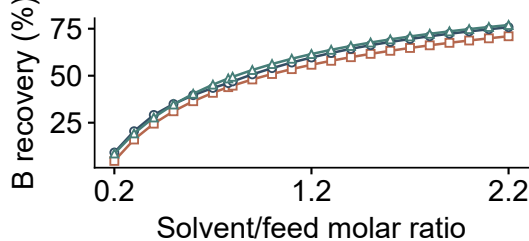
j Allocation: B recovery



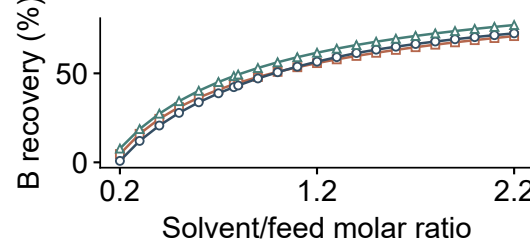
k Allocation: H retention



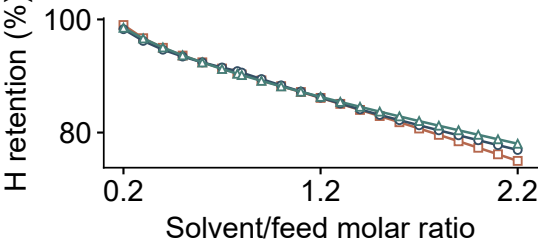
l Single stage: 298.15 K



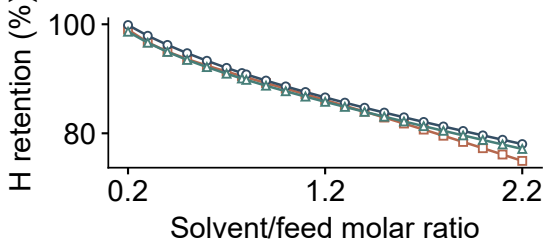
m Single stage: 303.15 K



n Single stage: 298.15 K



o Single stage: 303.15 K



—□— ThermoFormer —○— Experimental reference —△— UNIFAC-LLE